# Some Applications of Trigonometry (Set B) — MCQ Worksheet

**Mathematics** • Chapter 9 • *25 Questions, 1 mark each — Total: 25 marks*

***Instructions:*** *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

**Q1.** An angle measured from the horizontal looking up is the angle of:

(A) Depression
(B) Elevation
(C) Incidence
(D) Reflection

**Q2.** A tower casts a shadow equal to its height when sun's elevation is:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

**Q3.** Height of a building is 50 m. Length of its shadow when sun's elevation is 30° is:

(A) 50 m
(B) $50\sqrt{3}$ m
(C) $50/\sqrt{3}$ m
(D) 100 m

**Q4.** A 1.5 m boy at 28.5 m from a 30 m tall building. Angle of elevation of top of building from his eye is:

(A) 30°
(B) 45°
(C) 60°
(D) 75°

**Q5.** Angle of depression of a stone from a 100 m cliff is 60°. Distance of stone from base of cliff is:

(A) $100\sqrt{3}$ m
(B) $100/\sqrt{3}$ m
(C) 100 m
(D) 50 m

**Q6.** A ladder makes 60° angle with horizontal and its foot is 5 m from the wall. Length of ladder is:

(A) 10 m
(B) $5\sqrt{3}$ m
(C) 5 m
(D) $10\sqrt{3}$ m

**Q7.** From the top of a 25 m high tower, angle of depression of foot of a pole is 45°. Distance of pole from foot of tower is:

(A) 25 m
(B) $25\sqrt{3}$ m
(C) 12.5 m
(D) 50 m

**Q8.** Angle of elevation of top of a hill is 60° from a point P, and 30° from another point Q on the same line. PQ = 100 m. Height of hill is:

(A) $50\sqrt{3}$ m
(B) $100/\sqrt{3}$ m
(C) $100\sqrt{3}$ m
(D) 150 m

**Q9.** A kite is flying at height 100 m. String makes 30° with horizontal. Length of string is:

(A) 200 m
(B) 100 m
(C) $100\sqrt{3}$ m
(D) $200/\sqrt{3}$ m

**Q10.** From the top of a building 20 m tall, angle of depression of a car is 30°. Distance of car from foot of building is:

(A) $20\sqrt{3}$ m
(B) 20 m
(C) $20/\sqrt{3}$ m
(D) 10 m

**Q11.** Sun's altitude is 60°. The shadow of a 30 m high pole is:

(A) $10\sqrt{3}$ m
(B) 30 m
(C) $30\sqrt{3}$ m
(D) 60 m

**Q12.** A boy 1.6 m tall sees the top of a flagpole at elevation 60° when 10 m away. Height of flagpole is:

(A) $10\sqrt{3} + 1.6$ m
(B) $10 + 1.6\sqrt{3}$ m
(C) $10\sqrt{3}$ m
(D) 10 m

**Q13.** Sun is at altitude 30°. The shadow of a 50 m high pole is:

(A) 50 m
(B) $50/\sqrt{3}$ m
(C) $50\sqrt{3}$ m
(D) 100 m

**Q14.** From a window 15 m high above the ground in a street, angle of elevation of top of opposite house is 30° and angle of depression of foot is 45°. Height of opposite house is:

(A) $15 + 15/\sqrt{3}$ m
(B) $15 + 5\sqrt{3}$ m
(C) 30 m
(D) 20 m

**Q15.** A man at 60 m from a tower observes its top at elevation 30°. Height of tower is:

(A) $20\sqrt{3}$ m
(B) 20 m
(C) $60\sqrt{3}$ m
(D) 30 m

**Q16.** Two ships are on either side of a lighthouse 100 m high. The angles of depression of the ships from top are 30° and 45°. Distance between ships is:

(A) $100(\sqrt{3} + 1)$ m
(B) $100\sqrt{3}$ m
(C) 100 m
(D) 200 m

**Q17.** Top of a vertical pole is observed at angles of elevation 30° and 60° from two points on the same side, 20 m apart. Height of pole is:

(A) $10\sqrt{3}$ m
(B) $20\sqrt{3}$ m
(C) 10 m
(D) 30 m

**Q18.** An aeroplane at height 3000 m observes two ships sailing on the same line, with angles of depression 45° and 60°. Distance between ships is:

(A) $3000(1 - 1/\sqrt{3})$ m
(B) 3000 m
(C) $3000\sqrt{3}$ m
(D) 1500 m

**Q19.** From the top of a 7 m building, angle of elevation of top of cable tower is 60° and angle of depression of its foot is 45°. Height of tower is:

(A) $7(1 + \sqrt{3})$ m
(B) $7\sqrt{3}$ m
(C) 14 m
(D) 7 m

**Q20.** A statue 1.6 m tall stands on top of a pedestal. From a point on ground, the angles of elevation of top of statue and top of pedestal are 60° and 45°. Height of pedestal is:

(A) $0.8(\sqrt{3} + 1)$ m
(B) $1.6\sqrt{3}$ m
(C) $1.6/\sqrt{3}$ m
(D) 0.8 m

**Q21.** An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of top of chimney from his eye is 45°. Height of chimney is:

(A) 30 m
(B) 28.5 m
(C) 31.5 m
(D) 29 m

**Q22.** A tower stands vertically on the ground. From a point 15 m away from the foot of the tower, angle of elevation of top is 60°. Height of tower is:

(A) $15\sqrt{3}$ m
(B) 15 m
(C) $15/\sqrt{3}$ m
(D) 30 m

**Q23.** If shadow length of a tower is $\sqrt{3}$ times its height, the sun's altitude is:

(A) 30°
(B) 45°
(C) 60°
(D) 90°

**Q24.** Angle of elevation of cloud from a point 60 m above a lake is 30° and angle of depression of its reflection in lake is 60°. Height of cloud above lake is:

(A) 120 m
(B) 60 m
(C) 30 m
(D) 90 m

**Q25.** Distance covered by a man in time the angle of elevation changes from 45° to 60° while he walks towards a 60 m high tower is:

(A) $60(1 - 1/\sqrt{3})$ m
(B) $60(\sqrt{3} - 1)$ m
(C) $60/\sqrt{3}$ m
(D) 60 m

— End of Worksheet —